

Code-Layout, Readability and Reusability

Date	30 June 2026
Team ID	XXXXXX
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Consistent indentation throughout the project.
2	Proper File Structure	Files and folders are logically organized	Yes	Proper folder structure (templates, static, services, uploads).
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Meaningful variable and function names used.
4	Function / Method Names	Functions are descriptively named	Yes	Functions are modular and descriptive.
5	Code Comments	Inline and block comments explain logic	Partial	Comments added where necessary.
6	Modular Design	Code is split into reusable functions/modules	Yes	Reusable modules for AI services and frontend components.
7	No Redundant Code	Duplicate or unused code is removed	Yes	No unnecessary duplicate code.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Error handling implemented for API requests and file uploads.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Gemini Service	Python	Question Answering, Explanation	High
2	LaMini-Flan-T5 Module	Python, Transformers	Summarization	High
3	Frontend Template	HTML, CSS, JavaScript	All User Interfaces	High
4	FastAPI Routes	Python, FastAPI	All AI Features	High
5	File Upload Module	Python	PDF & Image Processing	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized modular project structure.
Readability	5	Clear naming conventions and clean code.
Reusability	5	Reusable services and frontend components.
Documentation / Comments	4	Basic comments included; can be expanded further.
Overall Score	4.8	Well-structured, maintainable, and reusable AI application.